

Adapter toward the y = 124 edge (y 63.5-123.5), CF column beside J2, ROM at the top edge

The diagram shows a top-down view of a PCB layout. Key features include:

- Dimensions:** A horizontal dimension of 170 mm is indicated at the top.
- Components:** Numerous integrated circuits (IC1-IC34), capacitors (C1-C30), resistors (R1-R14), and connectors (JP1, JP2, J2) are shown in their respective footprints.
- Annotations:**
 - "ADAPTER'S M20X0.6 HEADER (pin 1 above)" points to a red rectangular area labeled "J2".
 - "ADAPTER PIN 1 END" is marked near the bottom center.
 - A blue dashed box indicates a region with "H2 x 80.7 / y 87.5" and "power pads (approx.)".
 - A yellow dashed box on the right is labeled "ROM - keep clear, removable" and "keep-clear zone".
 - Other labels include "CF CARD INSERTS HERE (into the adapter)", "H1 x 80.7 / y 109.5", and "x1".
- Orientation:** The text "x (mm from the bus-conductor edge) ->" is at the top left, indicating the coordinate system.

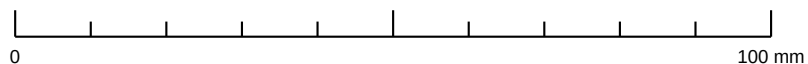
x from the bus edge, y from the JP1 end
H1 (pin-1 end): x 80.7 y 109.5
H2 (power-pad end): x 80.7 y 57.5
holes 52.0 apart, 3.2 mm drill, 7 mm copper keep-out
adapter x 65.4-109.4, y 53.5-113.5
header edge x 65.4, CF slot edge x 109.4
J2 pin 1: x 61.0 y 107.6 (square pad)
J2 pins y 59.4-107.6, odd row x 61.0, even x 58.4
ROM IC13 keep-clear x 154.3-175.9, y 34.8-89.0
(on the board: past an edge it is free air)

M3 hex 15 mm female-female (5.5 AF),
M3 x 6 screws + washers both sides
adapter underside 15 mm above the card

J2 pin 1 and the adapter's pin 1 at the SAME end (y max): stripe on that end at both plugs; the ribbon arches up from J2 and down into the adapter's header.

unrouted 0, vias 100, track 13748 mm
DRC copper violations: none

- ① J2 = the IDE header ON THIS BOARD (solid red, pin 1 = square pad); with its ribbon plug it stands ~18 mm; plug envelope dashed red
- ② SHORT 40-WIRE RIBBON, pin 1 to pin 1 (~5-8 cm between the plugs): the grey arrows, J2 -> the adapter's header
- ③ ADAPTER'S IDE HEADER (dashed blue): on the HX-2118P adapter, 15 mm above on standoffs H1 / H2 - the adapter does NOT plug into J2
- ④ CF CARD INSERTS HERE (into the adapter): the thick blue slot edge; the card is pushed in toward the bus



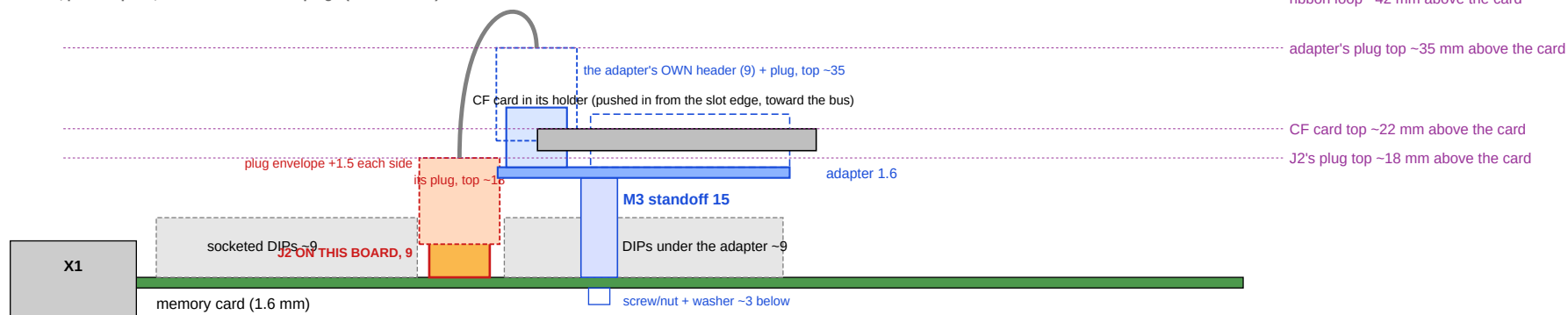
Also check with a ruler: the two blue hole crosses are 52.0 mm apart.

CHECK: this bar must measure exactly 100 mm - print at 100 % / "Actual size", NOT "Fit" / "Shrink to page"

Side view at 1:1 - the height stack (section along x, seen from the y-max end; card component side up)

Heights are typical figures, not measured: box header 9, IDC plug on it +9 (~18), plug body ~1.5 wider than the shroud each side, socketed DIP 8.5-9.5, adapter 1.6 thick, CF holder + card ~6-8 on the adapter.

T 40-WIRE RIBBON, pin 1 to pin 1, ~5-8 cm between the plugs (min. ~35 mm)



J2 is the IDE header ON THIS BOARD. The CF adapter does NOT plug into it: the adapter sits on two 15 mm M3 standoffs and has its OWN IDE header; a short 40-wire ribbon (~5-8 cm) joins the two, pin 1 to pin 1.

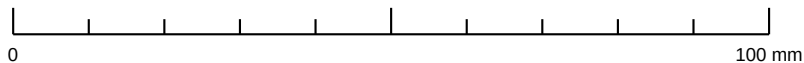
Standoffs: 15 mm M3 hex female-female (5.5 mm across flats) + M3 x 6 screws and washers (or 15 mm male-female with a nut under the card); 15 mm keeps ~3.5 mm between socketed DIPs (~9 mm) and the adapter's pin tails (~2-2.5 mm under it). J2 with its plug stands ~18 mm, taller than the standoffs: its plug envelope stays outside the adapter outline.

Total height above the card: CF card ~22 mm, the plug on the adapter's header ~35 mm, the ribbon loop ~42 mm; below the card: screw heads / nuts ~3 mm.

Slot pitch of the card cage: NOT KNOWN - if the neighbour card sits closer than ~44 mm above this card's component side, leave that slot empty.

Pin 20: the adapter has no pin 20 (key). J2 has it fitted: use a ribbon whose plugs have pin 20 OPEN, or pull J2's pin 20. JP2 (pin 20 = +5 V) stays OPEN.

Power: the adapter takes +5 V through its power pads, by a short cable from J3 (+5V, G, G, nc) - pad order to be confirmed on the adapter.



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